

‡ Sanguine Echoes — Combat Stats Reference

All base combat values pulled straight from `src/config/pLayerConfig.ts`, `enemyConfig.ts`, `gemConfig.ts` & `upgradeConfig.ts`. Images are the actual runtime spritesheets in `src/assets/`. Open this file from the repo root in a browser.

Tweak in config, not code. · Boss preset values are shown **before** the runtime `bossPreset()` ×3 HP/poise multiply.

🗡 The Ashen Hunter (Player) `src/config/playerConfig.ts`



Movement: 220 px/s ground · **jump** -520 v-up (1 air re-jump) · **gravity** 1300 · **fall cap** 600 px/s.

Dash (L): 260 px/s stamina-gated (100 pool, 25/dash, regen 30/s after 800ms). Ground = blanket i-frames; air = positional dodge with a 120ms contact-window i-frame.

Plunge (↓+J): dives at 950 px/s — **2.0×** base melee + 0.12 bonus/px fallen (<18), hard-capped 55. Bosses take 75% capped at 12% of max HP per hit.

Rally: 70% of damage taken becomes a recoverable orange pool (3s window, +1.2s per landed hit to a cap).

STAT	VALUE	DESCRIPTION
⚔ MELEE COMBAT		
attackDamage	25	Damage per standard melee hit.
attackRange	56	Hitbox reach in front of the hunter (px).
critChance	5%	Crit = 1.5× damage + big blood-red popup.
meleePoiseDamage	20	Poise chipped by a standard swing.
heavyPoiseDamage	35	Poise chipped by a charged heavy attack.
plungePoiseDamage	40	Poise chipped by the plunge (cap 15 post-crit).
parryPoiseDamage	40	Poise chipped by the parry riposte.
parryDamageMultiplier	1.5×	Riposte damage after a successful parry.
🪂 PLUNGE (↓+J)		
plungeSpeed	950	Downward dive speed (px/s).
plungeDamageMultiplier	2.0	Base vs a 1.0 standard swing.
plungeFallBonusPerPx	0.12	Bonus damage added per px fallen from dive start.
plungeMaxFallBonus	18	Ceiling on the fall-distance bonus.
MAX_PLUNGE_DAMAGE	55	Hard cap on base + fall bonus (pre-crit).
MAX_PLUNGE_HP_DAMAGE	55	Hard cap after a crit rolls.
MAX_PLUNGE_POSTURE_DAMAGE	15	Posture cap per hit — can't one-shot a stagger.
plungeAoEMultiplier	0.6	Landing shockwave AoE vs a standard swing.
BOSS_PLUNGE_RESISTANCE	0.75	Bosses take 75% plunge damage.
plungeBossMaxHpFraction	0.12	Bosses take at most 12% of max HP per plunge.
⚡ DASH & STAMINA		
dashSpeed / airDashSpeed	260	Dash travel speed (px/s).
dashDuration / airDashDuration	400ms	Ground = i-frames; air = 104px positional hover.
dashStaminaMax	100	Yellow stamina pool.
dashStaminaCost	25	Stamina consumed per dash.
dashRegenDelay / Rate	800ms / ±30/s	Regen after last dash.
dashContactIFrames	120ms	Air-dash i-frame window on threat contact.
🛡 DEFENSE, CHARGE & PARRY		
maxHardHealth	100	Blood-red hard HP pool.
rallyableFraction	0.7	Fraction of damage taken → recoverable Rally.

STAT	VALUE	DESCRIPTION
rallyDuration / RallyHitExtension	3000 / +1200ms	Rally window and per-hit extension (to cap).
chargeTime / ChargeQuickReleaseMs	800 / 150ms	Hold-to-charge heavy attack.
parryWindow	200ms	Timing window that negates a hit (K).
parryCooldown	800ms	Cooldown between parries.
parryHealFraction	5%	Successful parry restores this % of max HP (scales with Vitality).
hurtDuration / Invuln	180 / 400ms	Knockback state + hurt i-frames.
knockbackSpeed	260	Hurt knockback horizontal speed (px/s).
bodyWidth × bodyHeight	28 × 48	Collision box (matches ~30×48px art).
maxAirJumps	1	Air re-jumps after the ground jump.

 Echoing Surge (Ultimate) *src/config/playerConfig.ts · cast in playerStates.ts*

Key — U. A charge-driven mass-stagger + rally-conversion burst. The cyan meter below the HP bar fills over time (faster in combat); **when full, press u to cash all temporary Rally into permanent hard HP**, then the hunter is briefly invulnerable while a **teal shockwave** staggers, chips, and slows every enemy in range. The meter is spent back to zero and must be earned again. Crowd-clear, not a boss-killer.

STAT	VALUE	DESCRIPTION
 CHARGE		
ultimateChargeTimeMs	45,000ms	Base time to fill the meter fully (out of combat).
ultimateCombatChargeMultiplier	2×	Charge speed while in active combat — ~22.5s when a live enemy is threatening.
Combat detection pad	260×200	A live enemy within this AABB pad of the hunter (<nearbyEnemies) triggers the ×2 combat rate.
 SHOCKWAVE (ON CAST)		
ultimateRadius	240px	Spatial coverage of the surge from the hunter's origin.
ultimateDamage	60	Raw HP to every live enemy in range — crowd-clear, flat (no crit).
ultimatePoiseDamage	75	Mass poise stagger — breaks chaff outright, chips elites/bosses.
ultimateSlowDurationMs	2,500ms	How long a struck-but-surviving enemy stays slowed.
ultimateSlowFactor	0.45	Movement multiplier while slowed (45% of normal speed).
 CAST GRACE		
ultimateInvulnerabilityMs	600ms	Brief invuln while the surge resolves so you can't trade the cast for a hit.
ULTIMATE_CAST_MS	420ms	Surge pose duration (attack anim + flash) before control returns.
Rally to HP	100%	Cast converts every point of temporary Rally into permanent hard HP, then empties the pool.
Meter reset	→ 0%	The surge is spent — the HUD drops back to [U] 0% immediately.

 L1 — Gallows Graveyard Creeps *vampire · jumping-demon · flying-eye scout*

L1 CREEP · CHASER

Vampire



HP

75

Chase spd

95 px/s

Move spd

45

Damage

8

Detection

300

Attack rng

40

Poise

40

Stagger

800ms

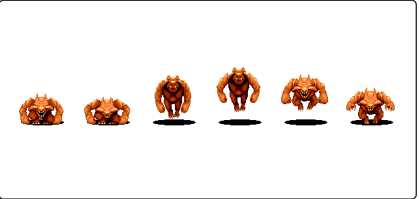
Echo

11

Body 70×80. Patrol ±120 of anchor.

L1 CREEP · POUNCER

Jumping Demon



HP

65

Chase spd

150 px/s

Move spd

60

Damage

10

Detection

350

Attack rng

50

Poise

40

Stagger

800ms

Echo

10

Body 60×70. Patrol ±160. Fastest chaser.

L1 SCOUT · FLYING

Flying Eye



HP

60

Chase spd

120 px/s

Move spd

70

Damage

6

Detection

380

Attack rng

45

Poise

30

Stagger

800ms

Echo

9

Body 44×40. **Flying** — gravity off, hovering patrol.

L2 — Haunted Forest

knight · spellcaster (wizard) · mutant-toad

L2 ELITE · ARMORED SLASHER

Terrible Knight



HP

130

Chase spd

85 px/s

Move spd

32

Damage

14

Detection

320

Attack rng

55

Poise

80

Stagger

1100ms

Echo

20

Attack frames

2–3

Cooldown

1500ms

Body 64×84. Sword slash: wind-up (safe) → frames 2–3 lethal sweep → recovery. Poise regen 40/s.

L2 CREEP · RANGED CASTER

Spellcaster (Wizard)



HP

70

Chase spd

70 px/s

Move spd

40

Damage

12

Detection

420

Attack rng

320

Poise

30

Stagger

800ms

Echo

11

Attack frames

2–3

Body 52×78. **Ranged** aimed magic bolt; body also lethal on the cast frames 2–3.

L2 CREEP · POUNCER

Mutant Toad



HP

90

Chase spd

115 px/s

Move spd

40

Damage

12

Detection

330

Attack rng

52

Poise

50

Stagger

850ms

Cooldown

2200ms

Echo

14

Body 60×42. Squat ground chaser that hops — patrol/chase loop mutant-toad-jump, melee short-hop lunge (mutant-toad-attack), idle reserved for stagger recovery + facingRight. Sheets: mutant-toad-idle/attack/jump.png (80×64).

🐾 **L3 — Night Town** berserker (werewolf) · caster (ghost) · fire-skull

L3 ELITE · BERSERKER
Werewolf Berserker



HP	150
Chase spd	150 px/s
Move spd	70
Damage	16
Detection	360
Attack rng	60
Poise	80
Stagger	1100ms
Echo	23

Body 56×66. **Super-armor while attacking** — can't be staggered mid-swing. Leap FSM: idle/run/jump-pounce/fall; art faces right; 900ms attack cooldown. Poise regen 40/s.

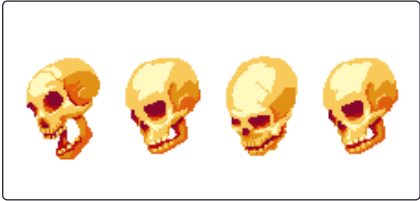
L3 CASTER · PHASE-SHIFTER
Ghost Caster



HP	80
Chase spd	60 px/s
Move spd	36
Damage	12
Detection	400
Attack rng	300
Poise	30
Stagger	800ms
Echo	12

Body 44×70. **Ranged** spectral bolt; phases out below half HP, re-materialises away.

L3 CASTER · FLOATING
Fire Skull

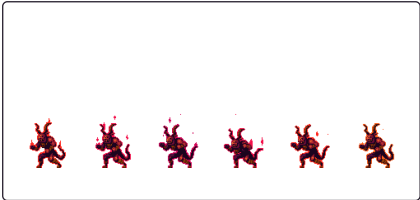


HP	80
Chase spd	65 px/s
Move spd	30
Damage	12
Detection	420
Attack rng	300
Poise	35
Stagger	850ms
Echo	14
Attack frames	2–3

Body 40×50. **Flying + ranged**: hovers like the eye, casts an exploding rotating fireball (dragon-lob logic); NoFire sheet idles, Fire sheet ignites on the cast frames 2–3.

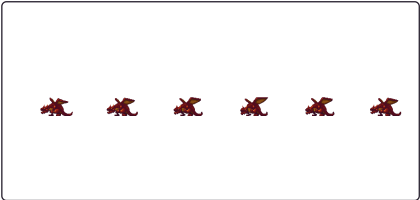
👹 **Bosses** config base values — runtime bossPreset() ×3 HP & poise

👑 L1 BOSS · HELL-BEAST
Hell Beast



HP base	260
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👑 L2 BOSS · HOLLOW WYRM
Dragon

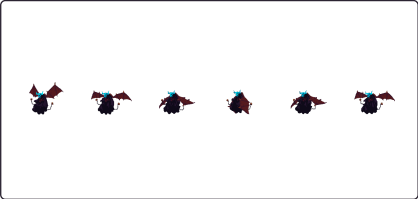


HP base	300
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HP in-game	780	HP in-game	900
Chase spd	90 px/s	Chase spd	100 px/s
Damage	20	Damage	22
Detection	520	Detection	500
Attack rng	230	Attack rng	130
Poise base	100	Poise base	120
Poise in-game	300	Poise in-game	360
Stagger	3500ms	Stagger	3500ms
Echo	39	Echo	51
Attack frames	—	Attack frames	3–5
Idle → Breath telegraph → Burn. Body 60×64. PBS 50/s.		144×64 art at 2× scale, faces left. Frames 3–5 of the breath = fire plume. Phase 2's only addition is the ranged fireball (no chase-speed or plume spike). Shortened ~4.5s post-stagger super-armor. Poise regen 75/s.	

👑 L3 BOSS · ARCH-DEMON (FINAL)

Arch Demon



HP base	420
HP in-game	1260
Chase spd	80 px/s
Damage	24
Detection	560
Attack rng	120
Poise base	180
Poise in-game	540
Stagger	3500ms
Echo	63
Attack frames	—

3 phases: breath + ember volley + magic bolt. Body 96×120. Poise regen 90/s.

💎 Temporary Upgrades — Blood Gems src/config/gemConfig.ts · rewarded for defeating a level boss (1-of-3 choice)

Run-long buffs chosen after each level boss (rolled 1-of-3). They modify a live-read player stat, so the FSM and combat pick the effect up **immediately**. Stored in `GameContext.runBuffs` and reset at the start of a new run — they do **not** persist to your next outing.

Mode — flat = the amount is **added** to the stat · mult = the stat is **multiplied** by the amount.

GEM	STAT	MODE	EFFECT
BOSS-REWARD CHOICE · TEMPORARY RUN BUFF			
Crimson Edge	attackDamage	mult	×1.2 — +20% attack damage.
Hollow Shard	critChance	flat	+8% critical strike chance.
Wraith Reach	attackRange	flat	+24 melee reach (px).
Blood Haste	moveSpeed	mult	×1.15 — +15% move speed.
Parry Knell	parryDamageMultiplier	flat	+0.5× parry riposte damage.
Rally Ember	rallyDuration	flat	+1500ms — rally window extended by 1.5s.

🔧 Permanent Upgrades — Sanctuary Tree src/config/upgradeConfig.ts · bought with banked Eldritch Echoes

Six upgrades bought with the **permanent** meta-currency Eldritch Echoes at the **Sanctuary** between levels. Ranks are folded into the hunter when a level is built and **persist** in `localStorage` across runs. Per-level echo cost climbs via `costGrowth` (a rank already owned costs more).

UPGRADE	STAT	BONUS / LEVEL	MAX	BASE COST	COST TO MAX
PER-LEVEL COST = ROUND(BASECOST × COSTGROWTH ^{LEVEL}) · COST TO MAX = SUMMED					

UPGRADE	STAT	BONUS / LEVEL	MAX	BASE COST	COST TO MAX
Vitality	maxHardHealth	+20 HP	5	50	791
Strength	attackDamage	+5	5	40	528
Reach	attackRange	+12	5	35	462
Riposte	parryDamageMultiplier	+0.25×	4	45	416
Endurance	rallyDuration	+500ms	5	40	528
Swiftness	moveSpeed	+5% (mult)	5	30	396

Sanguine Echoes · Combat Stats Reference · Mirrors `playerConfig.ts` / `enemyConfig.ts` / `gemConfig.ts` / `upgradeConfig.ts` · 2026-08-06 · Boss in-game HP/poise = $base \times BOSS_STAT_MULTIPLIER \ (3)$ · Upgrade cost-to-max = $\sum round(baseCost \times costGrowth^{level})$